
Education

- 2022/8 – 2027/5 (expected) **Ph.D. Linguistics, Purdue University**
Dissertation: Relative clause extraposition in English: a computational approach
Committee: Elaine J. Francis, Yan Cong, Arielle Borovsky, Thomas Stephan Juzek
- 2025 – present **M.S. Joint Statistics and Computer Science, Purdue University**
- 2019 – 2022 **M.A. Teaching Chinese to Speakers of Other Languages, Peking University, China**
- 2015 – 2019 **B.A. Chinese Linguistics and Literature & Finance, University of International Business and Economics, China**

Areas of Interests

- Broad Psycholinguistics, Computational Linguistics, Chinese as a Second Language Acquisition
- Narrow bilingual sentence processing, individual differences, syntactic ambiguity, syntactic alternation, automatic essay scoring, AI interpretability, multilingual evaluation, social media text analysis
- Methods experimental methods (acceptability judgement, self-paced reading, eye-tracking)
computational modeling (natural language processing, machine learning, language models)

Publications

Book Chapter

- In press Nie, D., Liu, Q., Jiang, D. Y., Yan, R., Zhang, Y., Fang, X., **Hu, J.**, Huang, X., & Yun, J. [Human reviewers in an AI era: A methodological reflection on research synthesis](#) In S. W. Chong (Ed.), *Research Synthesis Methodologies in Applied Linguistics*. pp. 330–347. John Benjamins.

Peer-reviewed Journal Articles

- 2026 **Hu, J.**, & Cong, Y. [LLM-derived metrics in second language writing assessment: an explainable AI approach](#). *Computers & Education*, 105721.
- 2026 Cong, Y., **Hu, J.**, Reese, T., & Liu, H. [How robust are linguistic markers of aging? The case of aging-related social media text](#). *Natural Language Processing Journal*. 14, 100203.
- 2023 Wang, Y., **Hu, J.**, An, Z., Li, C., & Zhao, Y. [The influence of metacognition monitoring on L2 Chinese audiovisual reading comprehension](#). *Frontiers in Psychology*. 14, 1133003.

Peer-reviewed Conference Proceedings

- Accepted **Hu, J.** & Issa, E. Where does the relative clause attach? Probing prosodic and semantic sensitivity in large audio language models. *The 3rd Conference on Language Modeling (COLM)*.
- 2025 **Hu, J.**, & Cong, Y. [Modeling Chinese L2 writing development: The LLM-surprisal perspective](#). *Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics (CMCL, held in conjunction with NAACL 2025)*. pp. 172-183. Association for Computational Linguistics (ACL).

- 17 **Hu, J.**, Cheng, Y.-F., & Bui, M. (2026, April 24–25). *From exploratory data analysis to predictive modeling: A multi-stage data science workflow for second language speaking proficiency modelling*. Poster presented at Purdue Linguistics Symposium (PLS), Purdue University, United States.
- 16 Alladin, M. A., Naik, S., **Hu, J.**, & Liu, C. (2026, April 24). *Knowledge graph construction from domain science papers using structure-aware and hierarchical RAG chunking*. Poster presented at Midwest Data Day (MDD), University of Chicago, United States.
- 15 **Hu, J.**, & Cong, Y. (2026, April 14–15). *LLM-derived metrics in L2 writing assessment: an explainable AI approach*. Poster presented at Purdue AI Research Showcase, Purdue University, United States.
- 14 **Hu, J.**, Cong, Y., & Francis, E. J. (2026, March 26–28). *Relative clause extraposition in English: An information-theoretic approach*. Poster presented at the 39th Annual Conference on Human Sentence Processing (HSP), MIT, United States.
- 13 **Hu, J.**, Francis, E. J., & Fang, S. (2026, March 26–28). *Temporal processing in L1 and L2 Chinese: A visual world eye-tracking study*. Poster presented at the 39th Annual Conference on Human Sentence Processing (HSP), MIT, United States.
- 12 **Hu, J.**, & Olson, D. J. (2026, March 21–24). *L2 proficiency and switch costs in bilingual sentence processing: Evidence from eye-tracking data*. Talk presented at the 2026 American Association for Applied Linguistics (AAAL) Conference, Chicago, Illinois, United States.
- 11 Fang, S., **Hu, J.**, & Francis, E. J. (2026, Jan 8–11). *Eye-tracking evidence on L2 scope interpretation: an approximate replication*. Talk presented at the 2026 Linguistic Society of America (LSA) Annual Meeting, New Orleans, United States.
- 10 **Hu, J.**, Francis, E. J., & Fang, S. (2025, Nov 6–9). *Integrative and predictive mechanisms in L2 processing of temporal information: Evidence from web-based visual world eye-tracking*. Poster presented at the 50th annual Boston University Conference on Language Development (BUCLD), Boston University, United States.
- 9 **Hu, J.**, Francis, E. J., & Fang, S. (2025, May 22–23). *Online processing of temporal information in Mandarin Chinese L2 learners: The role of individual differences*. Poster presented at the Language Learning roundtable (CogNeuroIDALL), University of Illinois Chicago, United States.
- 8 **Hu, J.**, Cong, Y., & Francis, E. J. (2025, May 7). *Leveraging AI language models for automatic writing assessment in Chinese as a second language*. Poster presented at the Office of Interdisciplinary Graduate Programs (OIGP) Spring Reception, Purdue University, United States.
- 7 **Hu, J.**, & Cong, Y. (2025, May 3–4). *Modeling Chinese L2 writing development: The LLM-surprisal perspective*. Poster presented at the Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL), Albuquerque, New Mexico, United States.
- 6 **Hu, J.**, Francis, E. J., & Fang, S. (2025, April 13–14). *Integration and prediction in online processing of temporal information: A study of L1 and L2 Chinese*. Talk presented at the Purdue Linguistics Symposium (PLS), Purdue University, United States.
- 5 **Hu, J.** & Cong, Y. (2025, April 4–6). *Leveraging AI language models for Chinese L2 writing assessment*. Talk presented at International Forum on Linguistics and Chinese Education (IFOLCE) Program & the 32nd International Conference on Chinese Language Instruction, Princeton University, United States.

- 4 **Hu, J.** & Zhao, Y. (2023, March 3–4). *The acquisition of the Chinese perfective marker -le by native speakers of English: A feature reassembly approach*. Talk presented at the 15th Annual Meeting of the Illinois Language and Linguistics Society (ILLS), University of Illinois Urbana-Champaign, United States.
- 3 Han, X., Hu, W., Xiang, Y., Ren, Y., & **Hu, J.** (2022, May 14–15). *Whether the separable word is a word or a phrase: An investigation based on the translation recognition of native speakers*. Talk presented at the 23rd Chinese Lexical Semantics Workshop (CLSW 2022, Best Paper Award), Fuzhou, China.
- 2 Han, X., Hu, W., Xiang, Y., Ren, Y., & **Hu, J.** (2021, May 30). *Whether the separable word is a word or a phrase: An investigation based on the translation recognition of native speakers*. Talk presented at the 29th “Challenge Cup” Wusi Youth Science Award Competition (First Prize), Peking University, China.
- 1 **Hu, J.** (2021, June 5). *The research trends of second language acquisition (2000–2020): Bibliometric analysis*. Talk presented at the 14th Academic Forum on Teaching Chinese as a Foreign Language, Peking University, China.

Professional Experience

2022/8 - present **Graduate Researcher**, [Purdue Experimental Linguistics Lab](#)

- Conducted psycholinguistic research on bilingual sentence processing using the visual world eye-tracking paradigm, investigating individual differences, event representation, and scope interpretation
- Analyzed large-scale eye-movement data using permutation cluster analysis and bootstrapped divergence point analysis
- Assisted in refining stimuli for another online self-paced listening experiment and managed participant recruitment, scheduling, and in-lab session coordination

2026/1 - 2026/5 **Graduate Research Assistant**, [Computational Linguistics @ Purdue Lab \(CL@P\)](#)

Project: Syntactic Ambiguity Resolution in Large Audio Language Models (LALMs)

- Designed and implemented an experiment on relative clause attachment to probe prosodic and semantic sensitivity in LALMs across multiple languages (English, Chinese, Arabic)
- Constructed prosodically controlled stimuli via text-to-speech synthesis and developed prompting strategies to probe model behavior, and conducted data analysis, visualization, and prepared manuscript

2024/8 - 2024/12 **Graduate Research Assistant**, [Purdue Computation and Linguistic Meaning Lab](#)

Project: Assessment of Cognitive Status of Older Social Media Users with Natural Language Processing

- Developed scalable NLP pipelines to preprocess 6,000+ social media posts
- Applied unsupervised learning methods (K-Means clustering) and dimensionality reduction (t-SNE) to identify and filter irrelevant data
- Computed sentiment, lexical & syntactic complexity, part of speech, and text similarity metrics as linguistic biomarkers to detect probable cognitive impairment
- Built predictive models to classify latent patients with cognitive impairment with healthy individuals

Teaching & Mentoring

Linguistics Courses

Spring 2026 **Teaching Assistant**, School of Interdisciplinary Studies, Purdue University
LING 398 *Computational Techniques and Programming for Linguists* ([Course Website](#))

Fall 2025 **Instructor of Record**, School of Interdisciplinary Studies, Purdue University
LING 201 *Introduction to Linguistics*

Chinese Language and Culture Courses

- 2022 - 2025 **Instructor of Record**, School of Languages and Cultures, Purdue University
CHNS 301 *Chinese Level V* (Fall 2023)
CHNS 202 *Chinese Level IV* (Spring 2023, Spring 2024)
CHNS 201 *Chinese Level III* (Fall 2023)
CHNS 102 *Chinese Level II* (Spring 2025)
CHNS 102 DIS *Chinese Level II* (Summer 2023, Summer 2024)
- 2020 - 2021 **Teaching Assistant**, School of Chinese as a Second Language, Peking University
Contemporary Chinese Lecture Series (Summer 2020)
Advanced Chinese (Summer 2020)
Intermediate Chinese (Spring 2021)
- Summer 2018 **Teaching Assistant**, University of International Business and Economics
Thinking Chinese: Nature and Culture in Early China

Mentorship

- Spring 2026 **Mentor**, Purdue Margo Katherine Wilke Undergraduate Research Internship Program
- Undergraduate researchers:
 - Alejandro Sebastian Gonzalez
 - Amy Kathleen Genz
 - Erin Rhyse O'Donnell
 - Hannah Grace Barsoum
 - Samuel Isaiah Piper
 - Tara Wong
 - Two projects presented at the Spring 2026 Undergraduate Research Conference (Literature, Linguistics, Languages, & Culture Colloquium) at Purdue University
 - *Effects of discourse-related factors on relative clause extraposition*
 - *The role of phrase length in relative clause extraposition*

Awards, Fellowships, & Grants

- 2026 **Bilsland Dissertation Fellowship**, Purdue University
- 2026 **Student Travel Award**, Annual Conference on Human Sentence Processing (\$400)
- 2025 **Paula Menyuk Award**, Boston University Conference on Language Development (\$400)
- 2025 **Conference Travel Funding**, School of Interdisciplinary Studies, Purdue University (\$1300)
- 2024 **Outstanding Achievement Teaching Award**, Purdue University
- 2023,2024,2025 **The PROMISE grant**, Purdue College of Liberal Arts (\$1000)
- 2021 **Scientific Research Excellency Scholarship**, Peking University
- 2020 **Academic Excellency Scholarship**, Peking University
- 2019 **Hongcai Scholarship**, Peking University
- 2019 **Outstanding Graduate Award**, Beijing Municipal Government, China

Service

- 2025 – present **Ad hoc Journal Reviewer**
 - *The Gerontologist*
 - *Chinese as a Second Language*
- 2026 **Program Committee**
 - 21st Workshop on Innovative Use of NLP for Building Educational Applications (BEA)
- 2023 – present **Conference Abstract Reviewer**
 - Second Language Research Forum (SLRF2025, SLRF2026)
 - Purdue Linguistics Symposium (PLS2023, PLS2024, PLS2025)

2025 – present **Grant Reviewer**, Purdue Graduate Student Government
2025 **Judge**, Purdue Undergraduate Research Conference
2023-2024 **Social Chair**, Purdue Linguistics Association

Skills

Programming R, Python, SQL, JavaScript
Statistical Tools SPSS, Stata, SAS (all used in coursework & projects)
Version Control Git
Text Analysis AntConc (quantitative text analysis); CiteSpace (bibliometric analysis)
Speech Analysis Praat (acoustic analysis)
Experiments Gorilla (eye-tracking experiment); PCIBex (self-paced reading); Qualtrics (online survey)
Languages Mandarin(native), Southern Min (native), English (proficient), Cantonese (intermediate)

Other Professional Development

June 2026 Video Research Institute (Purdue University)
June 2025 The North American Summer School for Logic, Language and Information (NASSLLI)
Oct – Nov 2024 The Penn State Workshops in Research Methods for Applied Linguist
Jan 2021 R Workshop for Linguistics (Hunan University, China)
July 2019 Advanced Seminar of Teaching Chinese as a Second Language (Peking University, China)